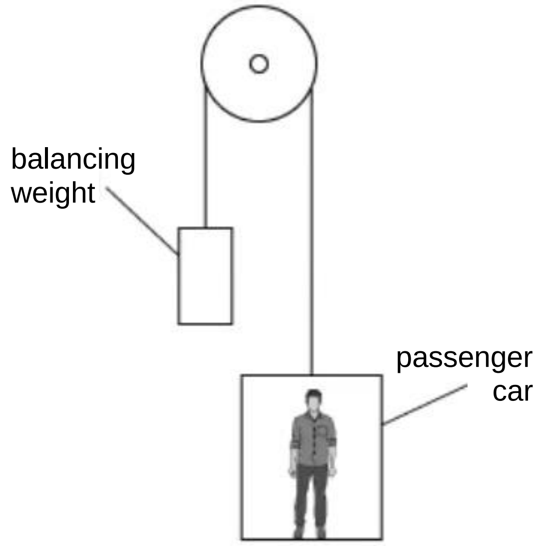
4 A lift consists of a passenger car supported by a cable which runs over a light, frictionless pulley to a balancing weight. The balancing weight falls as the passenger car rises.



(not drawn to scale)

The masses are shown in the table.

	mass / kg
passenger car	1200
balancing weight	1320
passenger	80

What is the magnitude of the force between the passenger and the passenger car when carrying just one passenger and when the pulley is free to rotate?

A 12 N **B** 773 N **C** 785 N **D** 797 N

4	**D**	for inextensible cable, the passenger, passenger car and balancing weight moves as one object with the upward motion of the balancing weight equivalent to the downward motion of the passenger and passenger car 1320 × 9.81 − (1200+80) × 9.81 = (1320+1200+80) a a = 0.15092 m s^{-2} resultant force on passenger = 80 × 0.15092 = 12.074 N upwards force by passenger car on passenger − weight of passenger = 12.074 N force by passenger car on passenger = 12.704 + 80 × 9.81 = 797 N

5 A heavy uniform beam of length L is supported by two vertical cords as shown